Number Systems and Data Representation 🔢

This section covers questions on converting numbers between different bases (decimal, binary, octal, hexadecimal) and character encoding schemes.

1. Converting the number 78210 to its base 5 equivalent would give? (A). 11135 (B). 32215 (C). 111125 (D). 101125 (E). 104115

- **Solution**: We use the method of successive division by the new base (5) and record the remainders.
  - 782÷5=156 remainder 2
  - 156÷5=31 remainder 1
  - 31÷5=6 remainder 1
  - 6÷5=1 remainder 1
  - 1÷5=0 remainder 1 Reading the remainders from bottom to top gives **11112**. The correct option is (C).

2. Converting the number 34425 to its decimal equivalent would give? (A). 64810 (B). 81810 (C). 71310 (D). 49710 (E). 41910

- **Solution**: We multiply each digit by its corresponding power of the base (5) and sum the results. 34425=(3×53)+(4×52)+(4×51)+(2×50) =(3×125)+(4×25)+(4×5)+(2×1) =375+100+20+2=49710. The correct option is (D).

3. Expressing the number ABC16 in base 7 would result in? (A). 110147 (B). 310147 (C). 1101047 (D) 110047 (E). 610417

- **Solution**: This is a two-step conversion: first from hexadecimal (base 16) to decimal (base 10), then from decimal to base 7. **Step 1: Base 16 to Base 10** In hexadecimal, A=10, B=11, C=12. ABC16=(10×162)+(11×161)+(12×160) =(10×256)+(11×16)+(12×1)=2560+176+12=274810. **Step 2: Base 10 to Base 7**
  - 2748÷7=392 remainder 4
  - 392÷7=56 remainder 0
  - 56÷7=8 remainder 0
  - 8÷7=1 remainder 1
  - 1÷7=0 remainder 1 Reading the remainders from bottom to top gives **11004**. The correct option is (D).

4. The coding scheme that is commonly used in IBM mainframes and can represent 256 characters is? (A). ASCII (B). Unicode (C). EBCDIC (D). BCD (E). XS-3 code

- **Solution**: **EBCDIC** (Extended Binary Coded Decimal Interchange Code) is an 8-bit character encoding used mainly on IBM mainframe and midrange computer operating systems. An 8-bit code allows for 28=256 different characters. The correct option is (C).

Computer Fundamentals & Hardware 🖥️

This section covers the basic components of computers, types of computers, and their historical development.

5. These computers are often used by Engineers, scientists and other professionals who need to run complex programs and often refer to personal computers connected to a network. (A). Super Computer (B) Workstation (C). Notebooks (D). Personal Digital Assistant (E). iPad Mini

- **Solution**: A **workstation** is a high-performance personal computer designed for technical or scientific applications. They are more powerful than standard PCs and are often connected to a network. The correct option is (B).

6. Which of the following retains its content until it is exposed to ultraviolet light? (A) PROM (B) EEPROM (C) DRAM (D) EPROM (E) none of the above

- **Solution**: **EPROM** stands for Erasable Programmable Read-Only Memory. Its contents can be erased by exposing the chip to a strong ultraviolet (UV) light source, allowing it to be reprogrammed. The correct option is (D).

7. Fetching, Decoding, Executing and Storing are functions performed by? (A) Registers (B) Control Unit (C) Arithmetic and Logical Unit (D) Input Unit (E) output unit

- **Solution**: The **Control Unit (CU)** is a component of the CPU that directs the operation of the processor. It performs the machine cycle, which consists of fetching an instruction from memory, decoding it, executing it, and storing the result. The correct option is (B).

Software & Programming 🧑‍💻

This part includes questions on programming languages, types of software, and programming concepts.

8. Which of these languages is considered as the first structured procedural language? (A). Oberon (B). MODULA-2 (C). SmallTalk (D). LISP (E). ALGOL

- **Solution**: **ALGOL** (Algorithmic Language) is considered the first high-level language to implement structured programming concepts like block structures (begin...end) and recursion, which heavily influenced later languages like Pascal and C. The correct option is (E).

9. A method for representing instructions in a module with a language very similar to a computer program code is also referred to as? (A). Flowchart (B). Algorithm (C). Logic (D). Pseudocode (E). Program

- **Solution**: **Pseudocode** is an informal, high-level description of the operating principle of a computer program or other algorithm. It uses the structural conventions of a normal programming language but is intended for human reading rather than machine reading. The correct option is (D).

Networking & The Internet 🌐

These questions cover protocols, communication channels, and the history of the internet.

10. The protocol that is commonly used when you download programs for your computer from Internet sites or when you upload webpages to a web server is known as? (A). Telnet (B) FTP (C). Gopher (D). TCP/IP (E). HTTP

- **Solution**: **FTP** (File Transfer Protocol) is a standard network protocol used for the transfer of computer files between a client and server on a computer network. The correct option is (B).

11. Which protocol allows a user's computer, once connected, to emulate a remote computer? (A). Telnet (B). FTP (C). Gopher (D). TCP/IP (E). HTTP

- **Solution**: **Telnet** is a network protocol that provides a command-line interface to communicate with a remote device or server, effectively allowing a user to control it as if they were physically present. The correct option is (A).

12. Which of the following has evolved to become what we know today as the internet? (A) Telnet (B) Intranet (C) world wide web (D) ARPAnet (E) Wide Area Networks

- **Solution**: The **ARPAnet** (Advanced Research Projects Agency Network) was an early packet-switching network and the first network to implement the TCP/IP protocol suite. Both technologies became the technical foundation of the Internet. The correct option is (D).

Sources